

EVIDENCE · CANVAS · PROSE

# math

Boardroom-quality math layouts for mathematicians, quants, ML researchers, physicists, statisticians, and economists. KaTeX-rendered equations with persona-appropriate surround.

## The closed-form estimator.

LATTICE · MATH

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $\hat{\beta}$  — OLS coefficient vector
- $X$  — design matrix,  $n \times p$
- $y$  — response vector, length  $n$
- $X^{\top} X$  — Gram matrix,  $p \times p$ , must be invertible

## The log-likelihood we maximise.

LATTICE · MATH

$$\ell(\beta) = \sum_{i=1}^n \left[ y_i \log \sigma(x_i^\top \beta) + (1 - y_i) \log(1 - \sigma(x_i^\top \beta)) \right]$$

WHERE

- $\ell$  — log-likelihood, concave in  $\beta$
- $\sigma$  — logistic link,  $\sigma(z) = 1/(1 + e^{-z})$
- $y_i$  — observed label,  $\in \{0, 1\}$
- $x_i$  — feature vector for observation  $i$

# Derivative of $f$ from first principles.

|                                                            |                                             |
|------------------------------------------------------------|---------------------------------------------|
| $f(x + h) = f(x) + f'(x) h + O(h^2)$                       | <i>Taylor expansion, <math>n = 2</math></i> |
| $f(x + h) - f(x) = f'(x) h + O(h^2)$                       | <i>subtract <math>f(x)</math></i>           |
| $\frac{f(x + h) - f(x)}{h} = f'(x) + O(h)$                 | <i>divide by <math>h \neq 0</math></i>      |
| $\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = f'(x)$ | <b>take the limit</b>                       |

# Intermediate Value Theorem.

**DEFINITION.** A function  $f : [a, b] \rightarrow \mathbb{R}$  is *continuous* on  $[a, b]$  if  $\lim_{x \rightarrow c} f(x) = f(c)$  for every  $c \in [a, b]$ .

**THEOREM.** Let  $f$  be continuous on  $[a, b]$  and let  $y$  lie strictly between  $f(a)$  and  $f(b)$ . Then there exists  $c \in (a, b)$  with  $f(c) = y$ .

**PROOF.** Set  $S = \{x \in [a, b] : f(x) < y\}$ .  $S$  is non-empty and bounded; let  $c = \sup S$ . Continuity at  $c$  forces  $f(c) = y$ .  $\square$

## Frequentist vs Bayesian point estimate.

### FREQUENTIST

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$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(y \mid \theta)$$

Maximises the likelihood — no prior. Uncertainty quantified by the sampling distribution of  $\hat{\theta}$  across hypothetical repeats.

### BAYESIAN

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$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid y)$$

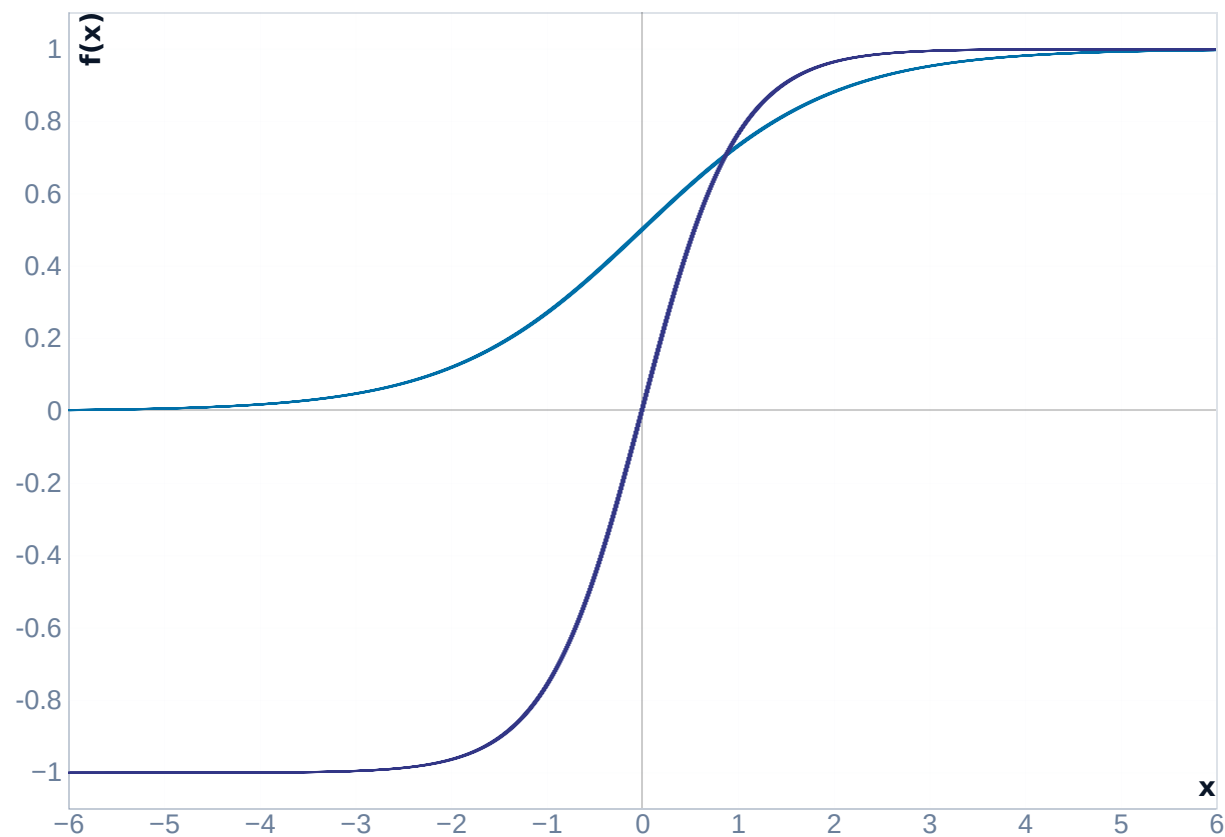
Maximises the posterior — conditions on the prior  $p(\theta)$ . Uncertainty is the posterior itself, no repeated sampling required.

# The sigmoid.

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$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Maps  $\mathbb{R} \rightarrow (0, 1)$ . *S*-shaped,  $\sigma(0) = 0.5$ ,  
steepest slope at the origin.



# The design matrix $X$ .

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$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

- `SHAPE` —  $n \times (p + 1)$
- `ROWS` — observations
- `COLS` — intercept +  $p$  features
- `RANK` — full-rank for OLS to have a unique solution
- `COLUMN 0` — all-ones, absorbs the intercept



## Effect of the treatment.

$$\hat{\beta} = 0.42 \pm 0.03$$

95% CI: [0.36, 0.48]

$p < 0.001$  ·  $n = 1,204$

*For every additional unit of exposure, the outcome rises by 0.42 SD — roughly an **8%** shift on the baseline. Effect size is the headline; the  $p$ -value just rules out chance.*

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## When NOT to reach for math.

**Two display equations in the base layout.** The bare math layout is built around one hero equation. For side-by-side display, use `math compare`. For a derivation chain, use `math derivation`. Stacking two `$$` blocks in the base layout breaks the visual contract.

**Symbols without a legend.** An equation with three undefined symbols is a puzzle, not a claim. Either every non-trivial symbol gets a legend entry, or the equation is simple enough that the audience knows it cold.

**ASCII math instead of KaTeX.** Writing `beta_hat = (X'X)^-1 X'y` as plain text bypasses the renderer. Always wrap math in `$$...$$` (display) or `$...$` (inline) — KaTeX is the entire reason this layout exists.

# See also.

## *RELATED COMPONENTS*

`code` — the implementation, not the equation, is the argument

`diagram` — the structure of the model, not its closed form

`stats` — a row of statistical results without a single equation focus

`content` — one inline equation inside a paragraph of prose